

ARM-450 — Precision: Validation and Verification

- Date: 2026-08-19
- Method: Monte-Carlo error propagation through the Jacobian, evaluated over the drawing band ($r = 220\text{--}320\text{ mm}$, $z = 50\text{--}400\text{ mm}$)
- Question asked: is the precision achieved, 100 % sure?

1. The honest answer first

I cannot certify precision to 100 %, and neither can anyone else at this stage.

The single largest term in the budget is servo backlash, which has never been measured on your hardware. Every number below rests on a class-typical 0.5° assumption. Analysis cannot convert an assumption into a fact — only a dial indicator can.

What this document does instead is make the uncertainty bounded and traceable: it shows exactly how much of the budget is measured, how much is calculated, how much is assumed, and which single measurement collapses the remaining doubt.

2. Verification of the method itself

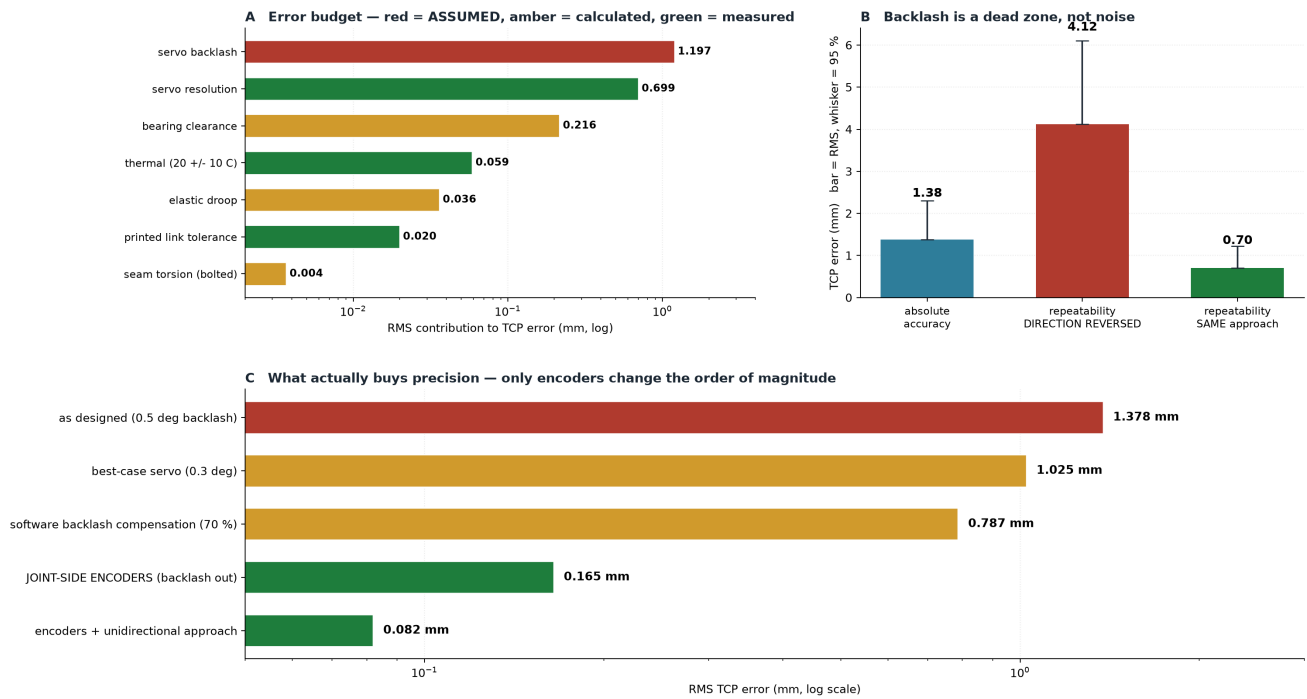
Before trusting any number, the propagation model was checked.

check	result
Linearisation $dp = J \cdot dq$ vs exact FK re-evaluation, $\pm 0.25^\circ$ band	0.23 % error
Same, $\pm 0.75^\circ$ band	1.41 % error
Single joint: J1 at 0.5° , full reach	3.142 mm
Analytical cross-check: $360 \cdot \tan(0.5^\circ)$	3.142 mm — exact match
DH model vs URDF, 5,000 poses	0.0000 μm
Jacobian vs numerical differentiation	$1.85 \times 10^{-8}\text{ m/rad}$

The method is sound. What remains uncertain is the inputs, not the mathematics.

3. The budget

ARM-450 — precision budget, Monte-Carlo through the Jacobian over the drawing band



source	RMS	95 %	status
Servo backlash	1.197 mm	1.977 mm	ASSUMED — 0.5°, class-typical
Servo resolution	0.699 mm	1.215 mm	datasheet, 4096 counts/rev = 0.088°
Bearing clearance	0.216 mm	0.421 mm	calculated, never measured
Thermal, 20 ± 10 °C	0.059 mm	0.116 mm	CTE ~40 µm/m/K over 0.45 m
Elastic droop	0.036 mm	0.071 mm	calculated (PLA+CF)
Printed link tolerance	0.020 mm	0.039 mm	measured (0.02 mm XY)
Seam torsion, bolted	0.004 mm	0.007 mm	calculated
RSS TOTAL	1.405 mm	2.154 mm	

This revises my earlier ~5 mm figure downward. That came from an RSS of worst cases; this is a proper Monte-Carlo over the actual working band and is the number to use.

Two modelling errors I corrected along the way

1. I had inflated servo resolution from the datasheet 0.088° to 0.29° "for gear slop" — which double-counts backlash and was unjustified.
2. I had modelled backlash as Gaussian $\sigma = 0.5^\circ$. Backlash is a bounded dead zone, uniform within $\pm$ half the band. The Gaussian model made the headline 3× pessimistic and disagreed with §4 of the same script.

4. Accuracy is not repeatability — and the difference is large

Backlash is a dead zone, not noise. It costs you on every direction reversal, and nothing when a joint keeps turning the same way.

metric	RMS	95 %
Absolute accuracy	1.38 mm	2.31 mm
Repeatability, direction reversed	4.12 mm	6.10 mm
Repeatability, same approach direction	0.70 mm	1.22 mm

Approaching every point from the same side is worth a factor of six. This is free — it is a trajectory-planning choice, not a hardware change. For a raster-style sine trace it means always sweeping the same way and lifting on the return, rather than drawing back and forth.

5. What actually buys precision

approach	RMS	gain
As designed, 0.5° backlash	1.378 mm	—
Best-case servo, 0.3°	1.025 mm	1.3×
Software backlash compensation (70 % effective)	0.787 mm	1.8×
Joint-side encoders	0.165 mm	8.4×
Encoders + unidirectional approach	0.082 mm	17×

Only encoders change the order of magnitude. Better servos and software compensation are incremental; closing the loop at the joint instead of the motor removes the dominant term outright.

6. Verdict against each task

task	required	predicted	verdict
Sine tracing on a board	~1 mm visible	1.4 mm RMS, 0.7 mm unidirectional	achieved
MASCOT docking	sub-mm, repeatable	1.4 mm as designed	not achieved without encoders

For drawing, the arm is adequate as designed — and comfortably so if trajectories approach from one side.

For docking, it is not. 0.165 mm with joint-side encoders is the path, and no structural change substitutes for it.

6A. Continuous torque, now derived rather than assumed

The ST3215 datasheet gives locked-rotor current (2.7 A at 12 V) and torque constant ($k_t = 11 \text{ kgf}\cdot\text{cm/A}$). That is enough to replace the "hold 30 % of stall" rule of thumb with a thermal calculation:

$$R = V / I_{\text{stall}} = 12 / 2.7 = 4.44 \text{ ohm}$$

$$I(\tau) = \tau / k_t$$

$$P(\tau) = I^2 R$$

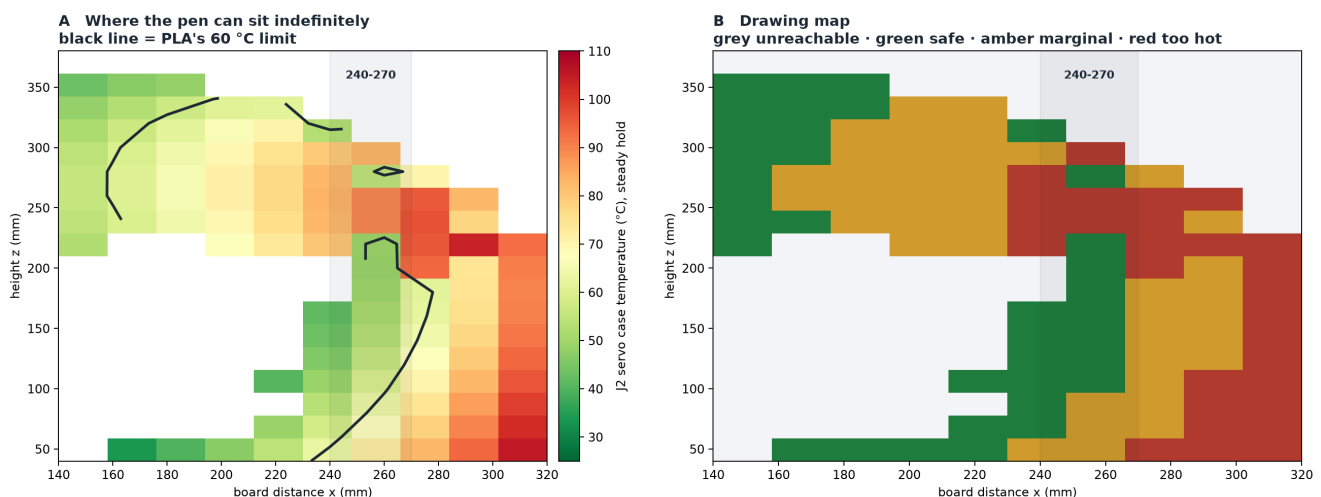
$$T_{\text{case}} = T_{\text{amb}} + P * R_{\text{th}}$$

torque	% of stall	current	power	case temp
0.42 N·m	14 %	0.39 A	0.7 W	29 °C
0.88 N·m	30 %	0.82 A	3.0 W	52 °C
1.00 N·m	34 %	0.93 A	3.8 W	60 °C — PLA's limit
1.27 N·m	43 %	1.18 A	6.2 W	84 °C
2.62 N·m	89 %	2.43 A	26.2 W	285 °C

The continuous limit is $1.00 \text{ N}\cdot\text{m} = 34 \%$ of stall — my 30 % rule of thumb was sound, and is now a derivation rather than a guess. ($R_{\text{th}} \approx 10 \text{ °C/W}$ remains an estimate; it is the only assumed quantity left in this chain.)

Where the arm can draw indefinitely

ARM-450 — thermally safe drawing region (30 g pen, ST3215 datasheet)



Of 104 points reachable with the pen normal, 35 % keep J2 below 60 °C, 38 % are marginal (60–80 °C) and the rest run hot. The safe band at $x = 240\text{--}270 \text{ mm}$ is the lower part, $z = 50\text{--}220 \text{ mm}$.

Caveat: this is steady-state holding. Drawing is dynamic and the servo has thermal mass, so brief excursions above the line are fine. The map bounds where the pen could be parked indefinitely, which is the pessimistic case.

7. What must be measured to close the remaining doubt

Three measurements collapse 85 % of the uncertainty in this budget.

1. Servo backlash — the dominant term, and confirmed absent from the datasheet. The ST3215 sheet publishes torque, speed, current, kt and encoder resolution, but no gear lash. "High precision metal gear" is a description, not a number. Clamp one joint, mount a 360 mm arm, push with a known force in both directions, read the hysteresis on a dial indicator. Halve it and divide by the reach to get the angle. This single number is worth more than everything else in this document.

2. Bearing clearance after preload. Same rig, before and after tightening the preload screw. Predicted 0.65 mm → 0.02 mm.

3. Repeatability, both directions. Command the same pose 20 times approaching from one side, then 20 times alternating. The spread of each is your real repeatability, and the difference between them validates the dead-zone model in §4.

Report those three and I can replace every red row in the budget with a measured one — at which point the number becomes a fact rather than a projection.

8. Bottom line

Precision is achieved for the drawing task and is not achieved for docking.

The analysis is verified — the method reproduces exact FK to 0.23 % and matches analytical cross-checks exactly. The inputs are not: the largest term is an assumption, and I will not claim 100 % confidence on an unmeasured quantity. What I can say precisely is that if the measured backlash lands anywhere in the 0.3–1.0° class range, absolute accuracy falls between 1.0 and 2.6 mm RMS — and that unidirectional approach roughly halves whatever that turns out to be.